



Course: Core Java

From Fundamentals to Object-Oriented Programming

Duration

- 60 Hours (Regular)
- 90 Hours (With Extensive Lab Practice)
- 120 Hours (Corporate & Placement-Oriented)

Prerequisites

- Basic Computer Knowledge
- Programming Logic (Preferred but Not Mandatory)

Module 1: Introduction to Java

Topics

- History of Java
- Features of Java
- Java Editions (SE, EE, ME)
- JDK, JRE, JVM
- Java Architecture
- Java Program Structure
- Installing Java and IDE Setup
- First Java Program

Lab Programs

- Hello World Program
- Basic Input and Output

Module 2: Java Fundamentals

Topics

- Variables
- Data Types
- Literals
- Type Casting
- Operators
- Expressions

Lab Programs

- Arithmetic Operations
- Simple Interest
- Area Calculations
- Temperature Conversion

Module 3: Control Statements



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Topics

- if Statement
- if-else
- Nested if
- else-if Ladder
- switch Statement

Lab Programs

- Largest Number
 - Grade Calculation
 - Menu-Driven Calculator
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Module 4: Looping Statements

Topics

- for Loop
- while Loop
- do-while Loop
- break Statement
- continue Statement

Lab Programs

- Factorial
 - Fibonacci Series
 - Prime Number
 - Armstrong Number
 - Pattern Programs
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Module 5: Arrays

Topics

- One-Dimensional Arrays
- Multi-Dimensional Arrays
- Array Operations

Lab Programs

- Sum of Array Elements
 - Largest Element
 - Matrix Addition
 - Matrix Multiplication
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Module 6: Strings

Topics

- String Class
- String Methods
- StringBuffer
- StringBuilder
- Immutable vs Mutable Objects

Lab Programs

- Reverse String
- Palindrome String
- Count Vowels
- Word Count

Module 7: Object-Oriented Programming Concepts

Topics

- Classes and Objects
- Constructors
- Instance Variables
- Methods
- Access Modifiers
- Encapsulation

Lab Programs

- Student Class
- Employee Class
- Product Class

Module 8: Inheritance

Topics

- Single Inheritance
- Multilevel Inheritance
- Hierarchical Inheritance
- Method Overriding
- super Keyword

Lab Programs

- Employee Management
- College Information System



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Module 9: Polymorphism

Topics

- Method Overloading
- Method Overriding
- Runtime Polymorphism
- Dynamic Method Dispatch

Lab Programs

- Calculator using Overloading
 - Shape Area Calculation
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Module 10: Abstraction

Topics

- Abstract Classes
- Abstract Methods
- Interfaces
- Multiple Inheritance using Interfaces

Lab Programs

- Banking Application
 - Payment System
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Module 11: Packages and Access Modifiers

Topics

- Packages
- Import Statement
- User-Defined Packages
- Access Specifiers

Lab Programs

- Package Creation
 - Package Utilization
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Module 12: Exception Handling

Topics

- Exception Hierarchy
- try
- catch



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- finally
- throw
- throws
- Custom Exceptions

Lab Programs

- Division by Zero
 - Bank Account Validation
 - Age Validation
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Module 13: Wrapper Classes

Topics

- Primitive Data Types
- Wrapper Classes
- Auto Boxing
- Auto Unboxing

Lab Programs

- Wrapper Conversion Examples
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Module 14: Collections Framework

Topics

List Interface

- ArrayList
- LinkedList
- Vector

Set Interface

- HashSet
- LinkedHashSet
- TreeSet

Queue Interface

- PriorityQueue

Map Interface

- HashMap
- LinkedHashMap
- TreeMap

Lab Programs

- Student Management System



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- Employee Data Processing
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Module 15: Generics

Topics

- Generic Classes
- Generic Methods
- Type Safety

Lab Programs

- Generic Collections
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Module 16: Multithreading

Topics

- Thread Lifecycle
- Creating Threads
- Runnable Interface
- Thread Class
- Synchronization
- Thread Priorities

Lab Programs

- Producer Consumer Example
 - Multi-threaded Counter
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Module 17: Java I/O Streams

Topics

- Byte Streams
- Character Streams
- Buffered Streams
- File Handling

Lab Programs

- File Reader
 - File Writer
 - Student Record Storage
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Module 18: Java 8 Features

Topics

- Lambda Expressions



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- Functional Interfaces
- Stream API
- Method References
- Optional Class
- Date and Time API

Lab Programs

- Employee Filtering
- Stream Operations

Module 19: JDBC (Java Database Connectivity)

Topics

- JDBC Architecture
- Drivers
- Connection Interface
- Statement
- PreparedStatement
- ResultSet

Lab Programs

- Insert Records
- Update Records
- Delete Records
- Search Records

Module 20: Mini Projects

Project 1

Student Management System

Project 2

Library Management System

Project 3

Employee Management System

Project 4

Banking Application

Project 5

Inventory Management System

Placement-Oriented Programming



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Topics

- Arrays
- Strings
- Collections
- OOP Interview Questions
- Exception Handling Questions
- Java Coding Challenges

Practice

- 100 Programming Problems
- 50 OOP Problems
- 25 Collection-Based Problems

Learning Outcomes

After completing this course, students will be able to:

- ✓ Develop Java Applications
- ✓ Apply Object-Oriented Programming Concepts
- ✓ Handle Exceptions and Multithreading
- ✓ Work with Collections Framework
- ✓ Connect Java Applications with Databases using JDBC
- ✓ Use Java 8 Features Effectively
- ✓ Build Real-Time Mini Projects
- ✓ Prepare for Java Developer Interviews and Campus Placements

Recommended Hands-on Practice

Activity	Count
Basic Programs	75
OOP Programs	50
Collection Programs	25
JDBC Programs	15
Assignments	10
Mini Projects	5
Assessments	5

Total Programs

150+ Hands-on Programs



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Next Course After Core Java

- Advanced Java (Servlets, JSP, JDBC Advanced)
- Spring Framework
- Spring Boot & Microservices
- Java Full Stack Development